

Real-Life Task: Amazon Laptop Market Analysis

TASK REQUIREMENTS

◇ Part 1: Web Scraping

Scrape laptop data from Amazon (minimum 100 products).

Extract the following fields:

1. product_title
2. Brand
3. RAM
4. Storage SSD
5. Window like : 10,11,12 ect
6. Color
7. Price
8. Rating (0 to 5)
9. Processor
10. Optionals (u can add)

product_title

```
title = soup.find("h2").get_text(strip=True) if soup.find("h2") else
None
```

brand (from title)

```
brand = re.match(r"^\w+", title).group() if title else None
```

price

```
price = re.search(r"₹\s?([\d,]+)", page_text).group(1).replace(",",  
"") if re.search(r"₹\s?([\d,]+)", page_text) else None
```

rating

```
rating = re.search(r"(\d+(\.\d+)?)\s?out of 5", page_text)  
rating = rating.group(1) if rating else None
```

processor

```
processor = re.search(r"(i[3579]|Ryzen\s?\d)", page_text)  
processor = processor.group() if processor else None
```

RAM

```
ram = re.search(r"(\d+)\s?GB\s?(RAM)?", page_text)  
ram = ram.group(1)+"GB" if ram else None
```

storage

```
storage = re.search(r"(\d+)\s?(GB|TB)\s?(SSD|HDD)?", page_text)  
storage = storage.group() if storage else None
```

screen_size

```
screen = re.search(r"(\d+(\.\d+)?)\s?(Inch|Inches)", page_text)  
screen = screen.group(1) if screen else None
```

Colour

```
color = re.search(r"(?:Colour|Color)\s*[:~]?s*([A-Za-z ]+)",
page_text) color = color.group(1).strip() if color else None
```

Window

```
# Try: "Windows 11", "Windows 10", "Win11", "Win10", etc.
windows_match = re.search(r'(?:Windows\s*|Win\s*)(\d+)', title_text,
re.IGNORECASE)
if windows_match:
    windows_version = "Windows " + windows_match.group(1)
```

◇ Part 2: DataFrame Creation

- Create a Pandas DataFrame
- Check shape, info, missing values
- Remove duplicates
- Convert price columns to numeric
- Extract brand from title if not directly available

Part 3: Save Files

Students must generate:

1. amazon_laptop_raw.csv
2. amazon_laptop_cleaned.csv



Part 4: Analysis Questions (Mandatory)

Students must answer:

1. Which brand has highest average price?
2. Which brand has highest average rating?
3. Top 5 most reviewed laptops?
4. What is the average discount percentage?
5. Price vs Rating correlation value?
6. Distribution of laptops by RAM (8GB, 16GB, etc.)
7. Which screen size is most common?
8. How many laptops under ₹50,000?

Expected Folder Structure (Submission)

```
Amazon_Laptop_Project/  
|  
├─ amazon_laptop_raw.csv  
├─ amazon_laptop_cleaned.csv  
├─ amazon_scraping.ipynb  
└─ analysis_report.pdf
```